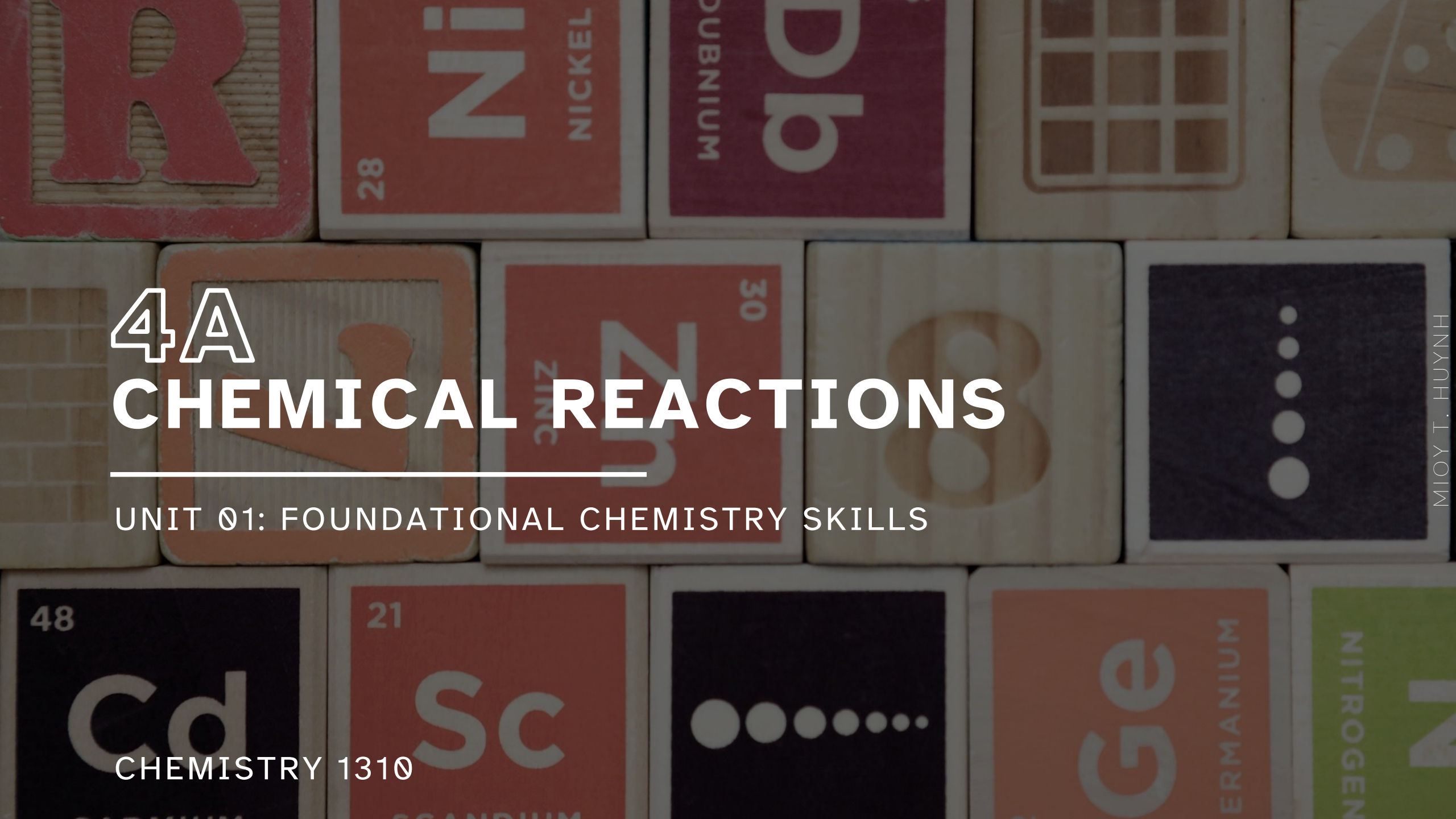


The background consists of a grid of wooden blocks, each representing a chemical element. Visible elements include Nickel (Ni, 28), Dubnium (Db), Zinc (Zn, 30), Cadmium (Cd, 48), Scandium (Sc, 21), Germanium (Ge), and Nitrogen (N). Some blocks also feature symbols like 'R', 'q', '8', and '3'.

# 4A CHEMICAL REACTIONS

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UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS

CHEMISTRY 1310

# LEARNING OBJECTIVES

## CHAPTER 4

1. Write complete and balanced chemical equations, based on either chemical symbols or word descriptions, including appropriate phases.
2. Recognize and describe the five basic types of chemical reactions.
3. List and describe the driving forces for reactions in aqueous solution.
4. Describe and write equations representing the dissociation of ionic compounds upon dissolution in water.
5. Define and apply the terms strong electrolyte, weak electrolyte, and nonelectrolyte.
6. Identify compounds as being strong electrolytes, weak electrolytes, or nonelectrolytes from chemical formulas.
7. Apply solubility guidelines to predict the formation of a precipitate in reactions of ionic compounds in aqueous solution.
8. Write and interpret ionic and net ionic equations for precipitation reactions.

# Chemical Equations

4.1

## Chemical equations

- Designed to represent the transformation of one or more chemical species into new substances.

Reactants: The starting components of a reaction

Products: The ending components.

Chemical equations are always written:

**Reactants → Products**

# Chemical Equations

## 4.1

A great deal of information can be gleaned from a chemical equation:

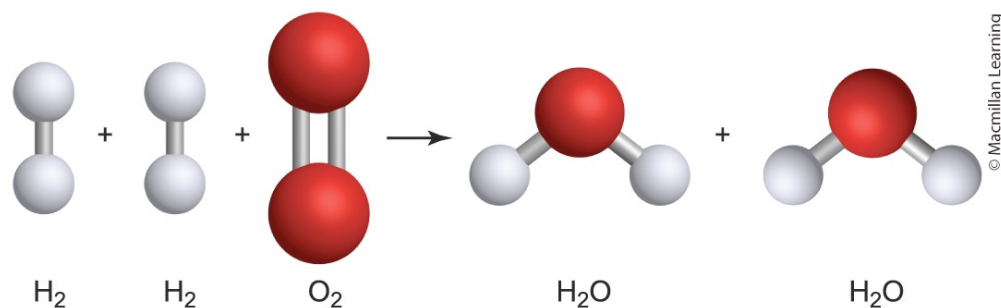


FIGURE 4.1

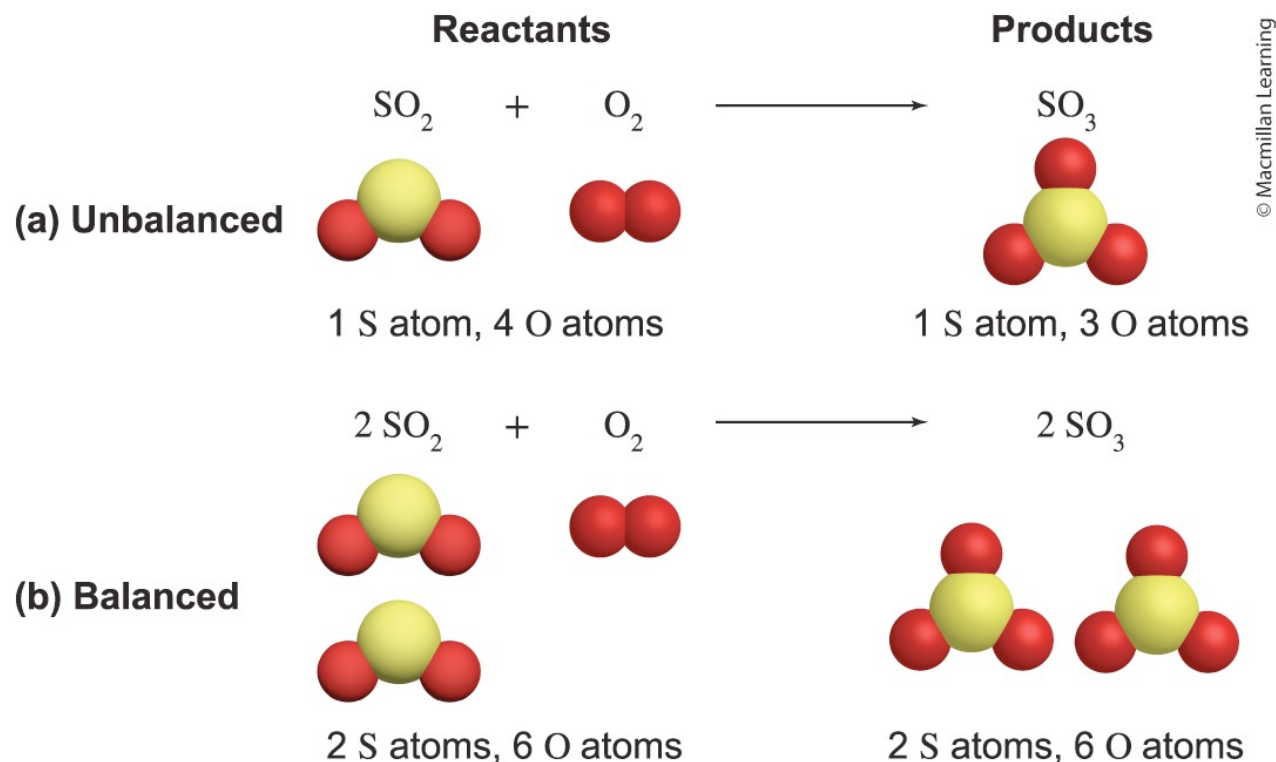
TABLE 4.1

| Information              | Notation                                                   | Example                                                                                             |
|--------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| What happens?            | Identity (names or formulas) of the reactants and products | Hydrogen ( $\text{H}_2$ ) and oxygen ( $\text{O}_2$ ) react to form water ( $\text{H}_2\text{O}$ ). |
| In what proportions?     | Coefficients placed before the formulas                    | $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$                                          |
| In what physical states? | (s), (l), (g), (aq) included after the formulas            | $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$            |
| Special conditions?      | Special conditions written above or below the arrow        | $\xrightarrow{\text{Heat}}$                                                                         |

# Chemical Equations

## 4.1

In a balanced chemical equation, the number of atoms of *each element* on the left (reactants) side of the chemical equation is equal to the number of atoms of that element on the right (products) side of the chemical equation.



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**FIGURE 4.3**

# Chemical Equations

## 4.1

Chemical equations can be balanced using this four-step strategy:

1. Balance polyatomic ions as whole units if they are present as both reactants and products.
2. Then, balance elements that appear in only one reactant and one product.
3. Then, balance any remaining elements; resolve odd/even issues by doubling previously determined coefficients.
4. Verify that the number of each type of atom is the same in the reactants and products and that the coefficients are in the simplest form (smallest whole number coefficients).

## EXAMPLE PROBLEM 1

Learning Objective(s): 4.1

Balance the reaction using the smallest set of whole number coefficients.



## EXAMPLE PROBLEM 2

Learning Objective(s): 4.1

Ammonia ( $\text{NH}_3$ ) reacts with oxygen gas to form water and nitrogen dioxide.

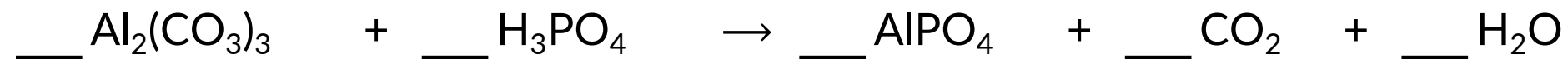
Write a balanced chemical equation using the smallest whole number coefficients.



## IN-CLASS QUESTION 1

Learning Objective(s): 4.1

What is the coefficient for each chemical species when the following reaction is balanced with the smallest possible whole number coefficients?



# Types of Chemical Reactions

## 4.2

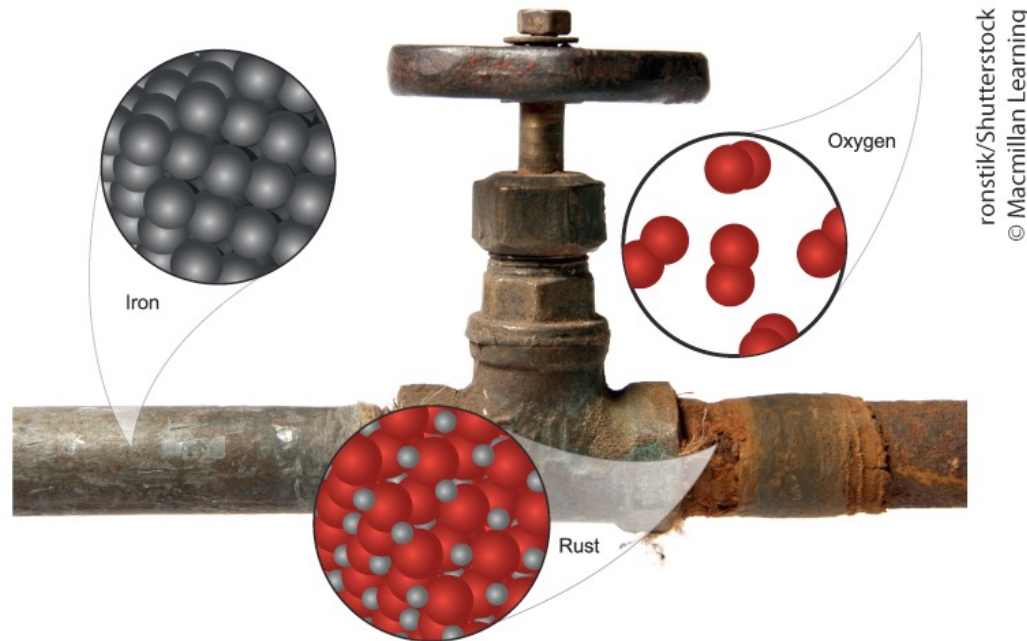
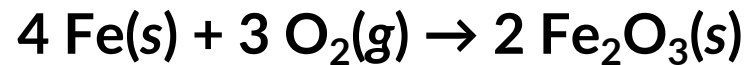
In this chapter, we'll explore five primary types of chemical reactions:

1. Synthesis reactions
2. Decomposition reactions
3. Single-replacement reactions
4. Double-replacement reactions
5. Combustion reactions

# Types of Chemical Reactions

## 4.2

Synthesis reactions (or combination reactions) start with simple reactants that combine to form a single more complex product.

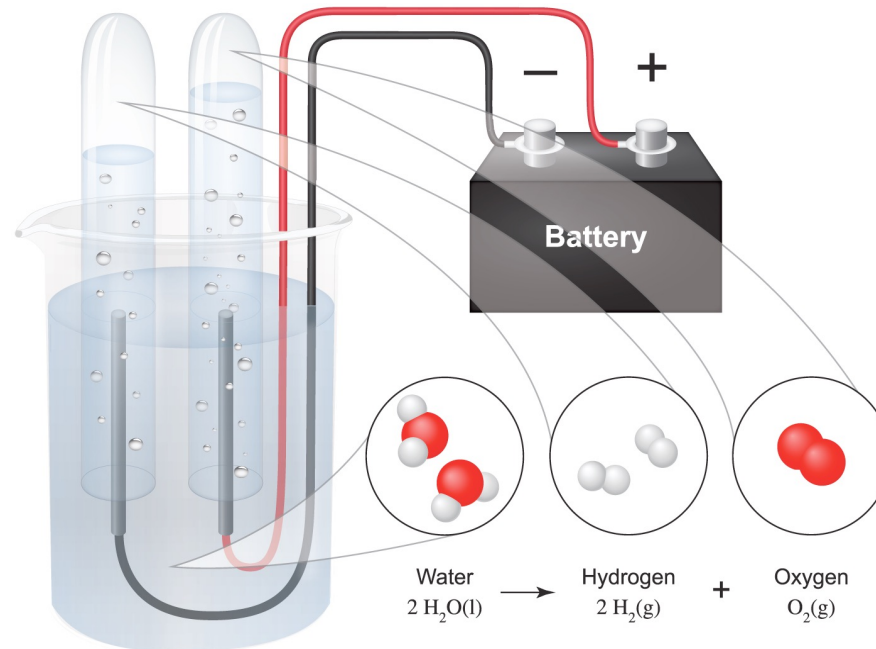
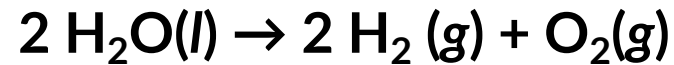


**FIGURE 4.6** Synthesis Reaction of Rust on Iron

# Types of Chemical Reactions

## 4.2

Decomposition reactions occur when a single reactant breaks down into less complex products.

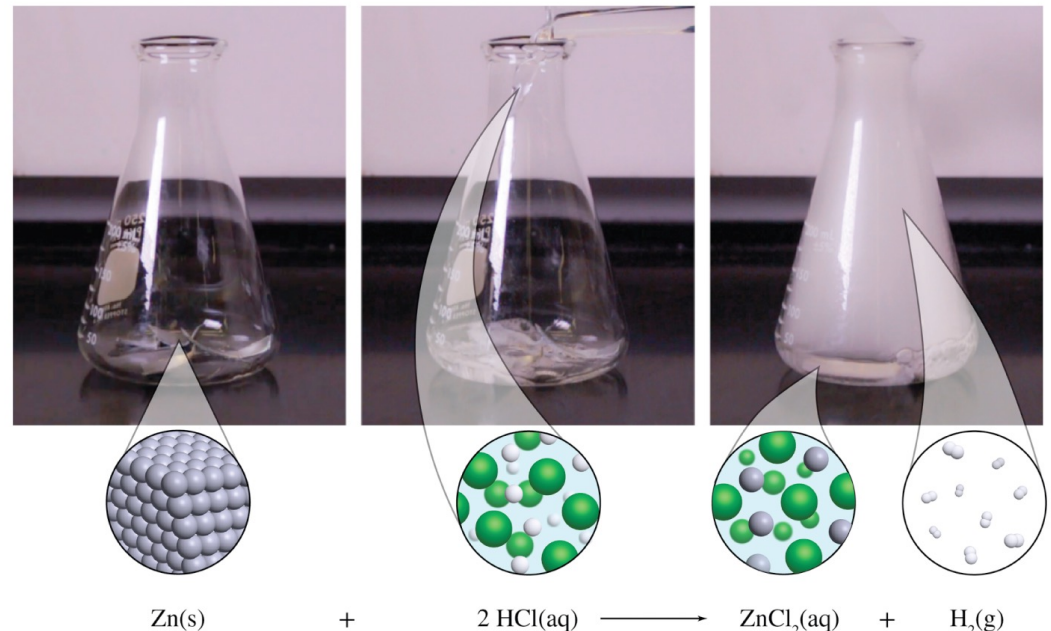
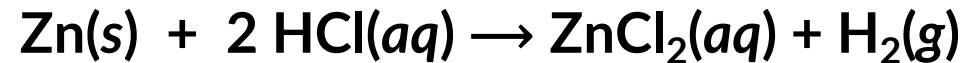


**FIGURE 4.7** Decomposition Reaction: Electrolysis of Water

# Types of Chemical Reactions

## 4.2

Single-replacement reactions (also called single-displacement reactions) occur when an element reacts with a compound and displaces one of the elements in that compound, producing a new compound and a new element.



**FIGURE 4.10** Single-Replacement Reaction of Zn with HCl

# Types of Chemical Reactions

## 4.2

Double-replacement reactions (also called double-displacement or metathesis reactions) occur when two ionic compounds exchange their constituent ions (cations and anions) to form two new ionic compounds.

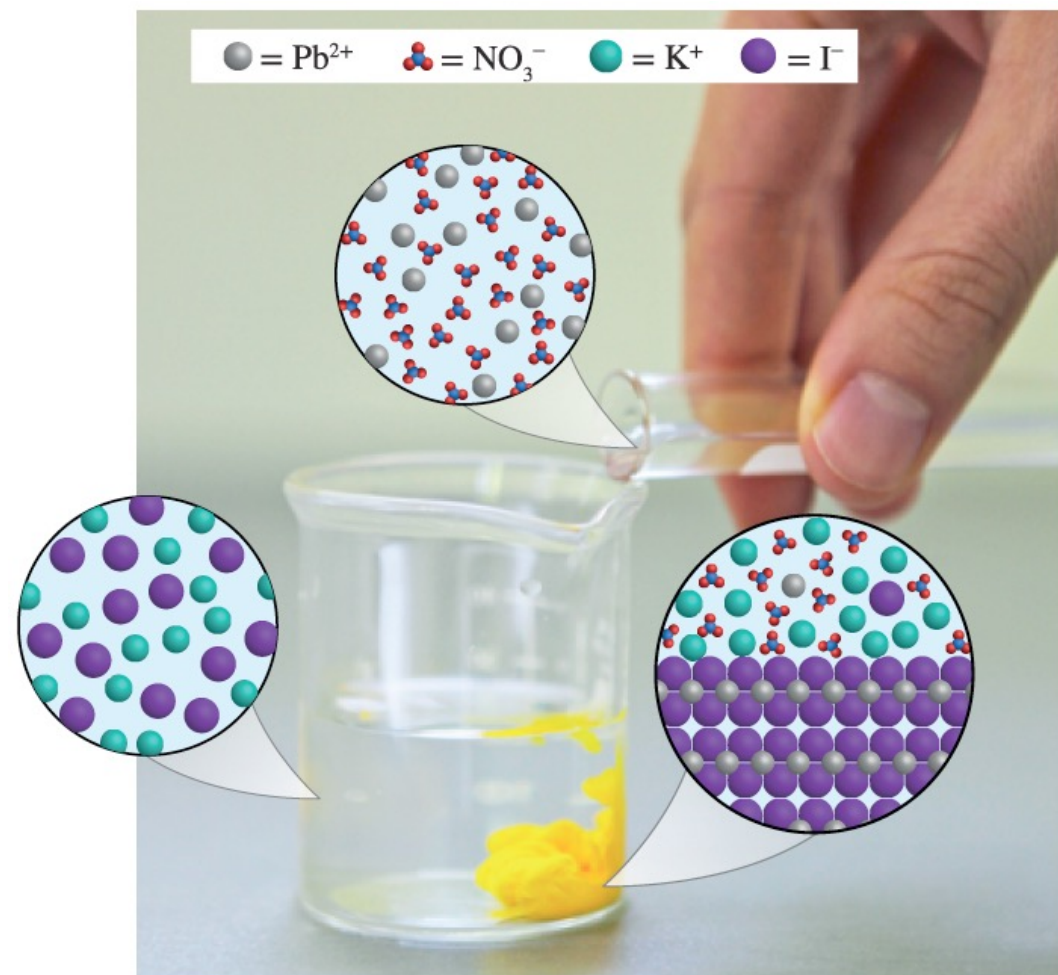
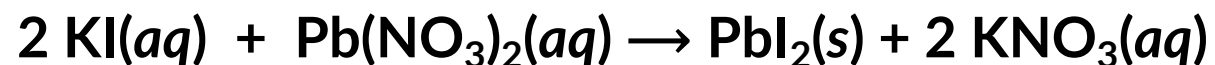


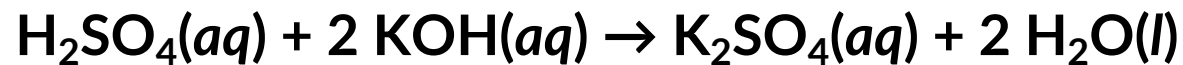
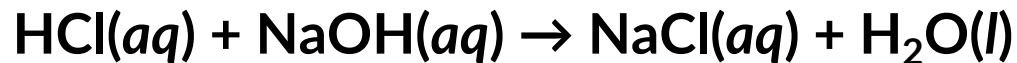
FIGURE 4.12 Double-Replacement Reaction of KI with  $\text{Pb}(\text{NO}_3)_2$

# Types of Chemical Reactions

## 4.2

Acid-base reactions are a special case of double-replacement reactions where water ( $\text{H}_2\text{O}$ ) is always a product.

- An acid is a type of compound usually written with H at the beginning of its formula.
- A base is a type of compound usually written with OH at the end of its formula (e.g. hydroxide compounds).

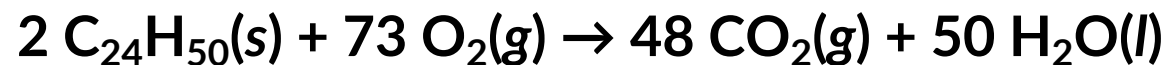
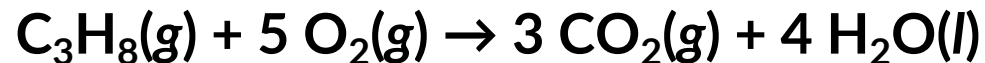


# Types of Chemical Reactions

## 4.2

Combustion reactions occur when a substance undergoes rapid combination with oxygen (O<sub>2</sub>).

- If the substance is a hydrocarbon (a molecular/covalent compound comprised of C, H, and sometimes O), then the products will always be carbon dioxide (CO<sub>2</sub>) and water (H<sub>2</sub>O).





# Types of Chemical Reactions

## 4.2

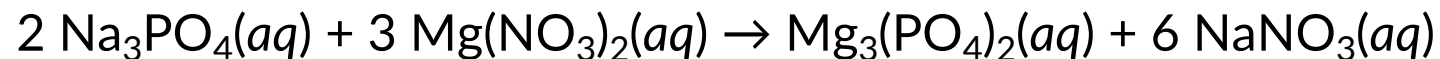
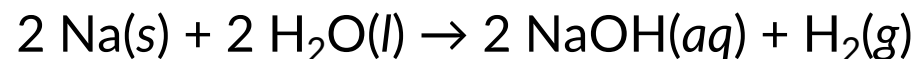
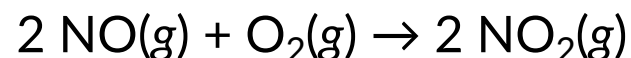
TABLE 4.2 Summary of Reaction Types

| Reaction Type      | Generic Formula                                                                  | Examples                                                                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Synthesis          | $A + B \rightarrow AB$                                                           | $4 \text{ Fe}(s) + 3 \text{ O}_2(g) \rightarrow 2 \text{ Fe}_2\text{O}_3(s)$                                                                                                                |
| Decomposition      | $AB \rightarrow A + B$                                                           | $2 \text{ H}_2\text{O}(l) \rightarrow 2 \text{ H}_2(g) + \text{O}_2(g)$<br>$2 \text{ KClO}_3(s) \rightarrow 2 \text{ KCl}(s) + 3 \text{ O}_2(g)$                                            |
| Single-Replacement | $A + BC \rightarrow AC + B$                                                      | $\text{Zn}(s) + 2 \text{ HCl}(aq) \rightarrow \text{ZnCl}_2(aq) + \text{H}_2(g)$                                                                                                            |
| Double-Replacement | $AB + CD \rightarrow AD + CB$                                                    | $2 \text{ KI}(aq) + \text{Pb}(\text{NO}_3)_2(aq) \rightarrow \text{PbI}_2(s) + 2 \text{ KNO}_3(aq)$<br>$\text{HBr}(aq) + \text{KOH}(aq) \rightarrow \text{KBr}(aq) + \text{H}_2\text{O}(l)$ |
| Combustion         | $\text{C}_x\text{H}_y + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ | $\text{C}_3\text{H}_8(g) + 5 \text{ O}_2(g) \rightarrow 3 \text{ CO}_2(g) + 4 \text{ H}_2\text{O}(l)$                                                                                       |

## IN-CLASS QUESTION 2

Learning Objective(s): 4.2

Label the type of chemical reaction using the terms: synthesis, decomposition, single-replacement, or double-replacement.



# Types of Chemical Reactions

## 4.2

Reactions occur spontaneously due to a combination of changes in heat energy and randomness, which are discussed in future chapters.

Certain patterns of reactants and products tend to be associated with spontaneous reactions and can be classified by their driving force:

1. Formation of a precipitate
2. Formation of water or neutralization (discussed in Chapter 16)
3. Oxidation and reduction (discussed in Chapter 19)

## IN-CLASS QUESTION 3

Learning Objective(s): 4.3

Identify the driving force for each reaction:

